

Customer Retention & Churn Analysis Report

Future Interns – Data Science & Analytics Track

Task 2: Customer Retention & Churn Analysis

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1. Introduction

Customer retention is a critical factor in the success of subscription-based businesses. Acquiring new customers is often more expensive than retaining existing ones, making churn analysis an essential business activity.

The objective of this project is to analyze customer behavior and identify factors that contribute to customer churn. Using customer subscription data, the analysis explores churn patterns, retention drivers, customer demographics, service usage, and payment preferences. The insights obtained can help businesses develop strategies to improve customer retention and reduce customer loss.

2. Dataset Overview

The analysis was performed using the Telco Customer Churn dataset containing customer demographic information, subscription details, service usage, billing information, and churn status.

Dataset Summary

- Total Records: 7,043 Customers
- Features: 21 Original Columns
- Target Variable: Churn
- Churned Customers: 1,869
- Active Customers: 5,174
- Overall Churn Rate: 26.54%

Key Attributes

- Customer Demographics

- Contract Type
 - Internet Service Type
 - Payment Method
 - Monthly Charges
 - Total Charges
 - Customer Tenure
 - Churn Status
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3. Data Cleaning and Preparation

The following preprocessing steps were performed:

Data Quality Checks

- Verified dataset structure and data types
- Checked for missing values
- Checked for duplicate records

Data Transformations

- Converted TotalCharges column to numeric format
- Handled missing values in TotalCharges
- Created Churn_Flag column for numerical analysis
- Created Tenure_Group categories:
 - 0–12 Months
 - 13–24 Months
 - 25–48 Months
 - 49–72 Months
- Created Monthly Charge groups for customer segmentation

The cleaned dataset was exported for Power BI dashboard development.

4. Churn Analysis

Overall Churn Performance

The analysis revealed that approximately 26.54% of customers have churned, indicating a significant opportunity for retention improvement.

Customer Distribution

- Active Customers: 73.46%
 - Churned Customers: 26.54%
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Contract-Based Churn Analysis

Contract type emerged as one of the strongest predictors of customer churn.

Contract Type	Churn Rate
Month-to-Month	42.71%
One Year	11.27%
Two Year	2.83%

Insight

Customers on month-to-month contracts are substantially more likely to leave compared to customers on long-term contracts.

Internet Service Analysis

Internet Service	Churn Rate
Fiber Optic	41.89%
DSL	18.96%
No Internet Service	7.40%

Insight

Fiber optic customers show the highest churn rates, suggesting possible issues related to pricing, service quality, or customer expectations.

Payment Method Analysis

Payment Method Churn Rate

Electronic Check 45.29%

Mailed Check 19.11%

Bank Transfer 16.71%

Credit Card 15.24%

Insight

Customers using electronic checks are significantly more likely to churn compared to customers using automated payment methods.

Tenure Analysis

Tenure Group Churn Rate

0–12 Months 47.68%

13–24 Months 28.71%

25–48 Months 20.39%

49–72 Months 9.51%

Insight

The highest churn occurs during the first year of subscription. Customers who remain beyond two years demonstrate significantly higher retention rates.

Demographic Analysis

Senior Citizen Status

Customer Type	Churn Rate
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Senior Citizens	41.68%
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Customer Type Churn Rate

Non-Senior Citizens 23.61%

Partner Status

Partner Status Churn Rate

No Partner 32.96%

Has Partner 19.66%

Dependents

Dependents Churn Rate

No Dependents 31.28%

Has Dependents 15.45%

Insight

Customers with stronger household commitments tend to remain customers longer and exhibit lower churn rates.

Monthly Charges Analysis

Customer Status Average Monthly Charges

Active Customers \$61.27

Churned Customers \$74.44

Insight

Customers paying higher monthly charges are more likely to churn, suggesting pricing sensitivity among customers.

5. Key Findings

The analysis identified several major drivers of customer churn:

1. Month-to-month contracts exhibit the highest churn rates.

2. Fiber optic customers churn more frequently than DSL customers.
 3. Electronic check users are the most likely customer group to leave.
 4. New customers within their first year are at the highest risk of churn.
 5. Senior citizens demonstrate significantly higher churn rates.
 6. Customers with higher monthly charges tend to churn more frequently.
 7. Long-term contracts strongly improve customer retention.
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6. Business Recommendations

Based on the findings, the following recommendations are proposed:

Improve Long-Term Contract Adoption

Encourage month-to-month customers to transition into one-year or two-year contracts through discounts and loyalty incentives.

Strengthen Early Customer Engagement

Develop onboarding programs targeting customers within their first 12 months to reduce early churn.

Review Fiber Optic Service Experience

Investigate pricing, customer support, and service quality concerns among fiber optic customers.

Promote Automated Payments

Encourage customers using electronic checks to switch to automated payment methods through incentives and simplified enrollment.

Monitor High-Risk Segments

Implement retention campaigns specifically targeting:

- Senior citizens
- New customers
- High monthly charge customers
- Month-to-month subscribers

7. Dashboard Overview

An interactive Power BI dashboard was developed to visualize customer churn patterns and retention drivers.

The dashboard includes:

- Customer KPIs
- Churn Rate by Contract Type
- Churn Rate by Internet Service
- Churn Rate by Payment Method
- Churn Rate by Tenure Group
- Churn Rate by Senior Citizen Status
- Churn Rate by Partner Status
- Churn Rate by Dependents
- Average Monthly Charges by Churn Status
- Interactive Slicers and Filters

Dashboard screenshots are included within the project repository.



